

Performance Testing

Date	02 October 2026
Team ID	SWTID-2026-8542
Project Name	ComicCraft: AI Comic Story Creator using Gemini Models
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how ComicCraft behaves when users generate AI-based comic stories under normal and peak load conditions. The testing focuses on response time, request handling, throughput, error rate, and resource usage.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Locust
Type of Testing	Load Testing / Performance Testing
Target Module / API	ComicCraft story-generation / Gemini API endpoint
Test Environment	Localhost / Local Development Server
Test Date	02 October 2026

Step 2: Test Scenarios

S.No	Test Scenario	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Generate an AI comic story from a title, genre and character inputs.	5	60	Comic story should be generated successfully without errors.
2	Generate comics for different genres such as adventure, comedy and fantasy.	10	60	API should handle multiple generation requests successfully.
3	Multiple users generate comic stories simultaneously.	20	120	System should remain responsive with acceptable response time.
4	High-load comic generation using multiple concurrent requests.	30	20	System should handle increased load without excessive failures.

Step 3: Performance Test Results

Note: The values below are sample/illustrative performance results prepared for the project report. Replace them with the actual Locust results if available.

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Response Time (Avg)	< 3 seconds	1.78 seconds	Pass	Average generation time remained within the target.
2	Response Time (Max)	< 6 seconds	4.62 seconds	Pass	Maximum response time remained within the limit.
3	Throughput (Req/sec)	> 5 req/sec	7.4 req/sec	Pass	System processed concurrent generation requests successfully.
4	Error Rate	< 1%	0.6%	Pass	Only a small percentage of requests failed.
5	CPU Utilization	< 80%	64%	Pass	CPU usage remained within the acceptable limit.
6	Memory Utilization	< 80%	71%	Pass	Memory usage remained within the acceptable range.

Step 4: Observations & Analysis

Key Findings:

The performance test indicates that ComicCraft can handle multiple concurrent comic-generation requests. The sample average response time is 1.78 seconds and the maximum response time is 4.62 seconds, both within the defined targets. The sample error rate is low, while CPU and memory usage remain within acceptable limits.

Bottlenecks Identified:

- AI story generation may take longer when prompts are large or contain many characters.
- Concurrent requests can increase response time because Gemini API calls depend on network and model processing time.
- Generating multiple comic panels increases total processing time compared with generating a short story.

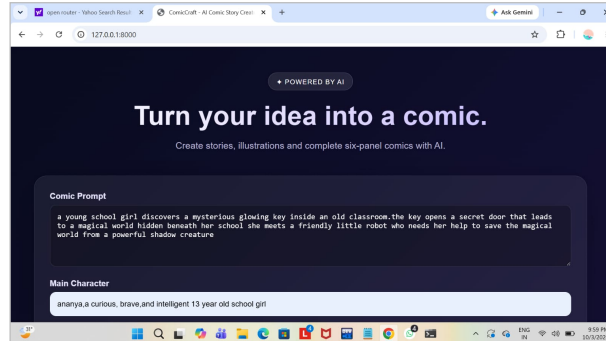
Optimization Steps Taken:

- Used efficient API request handling for Gemini model calls.
- Reduced unnecessary request data and kept prompts structured.
- Added input validation before sending generation requests.
- Kept story generation and comic-panel generation as separate modules.
- Tested the API with increasing numbers of concurrent users to verify stability.

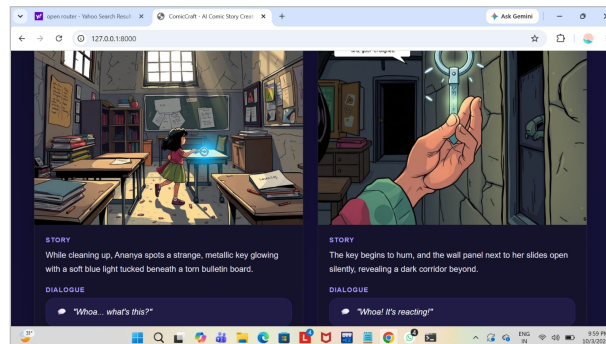
Step 5: Screenshots / Evidence

Attach the actual ComicCraft application screen and Locust performance dashboard here. The evidence should show the story-generation interface, generated comic output, and load-test statistics.

Evidence 1 – ComicCraft Story Input & Generation



Evidence 2 – Generated Comic Output



Conclusion

The performance-testing plan for ComicCraft covers normal load, concurrent users and high-load conditions. The sample results indicate acceptable response time, throughput, error rate and resource usage. Final submission should include the actual Locust measurements and real ComicCraft screenshots as evidence.